Midterm Super Doc

Real Estate Investment and Development | Lessons 5, 6 and 7

Built from the Lesson 5, 6 and 7 decks and the Midterm Review sheet. Every slide question is here with a worked answer, every calculation was re-run, and the three Midterm Review practice problems are solved in full.

Key idea

How to use it. Part 1 is the formula sheet. Parts 2 to 4 are the lessons: concepts, then calculations, then every in-class and practice question. Part 5 solves the three Midterm Review problems. Part 6 is a rapid-fire self-test.

Source tags. Slide numbers are in brackets, like [L6 s12]. Anything I added that is not on a slide is labeled "my answer" or "my note".

Lesson 4 is not in the files you sent. EBC, IBC, LTV and DSCR show up on the midterm. I define them in Part 1 from how the Lesson 6 slides use them. Check against your Lesson 4 notes.

Midterm coverage map

Midterm Review topic	Where it lives
L5 Concepts: market analysis steps, demand drivers by property type	Part 2, sections 2.6 and 2.7
L5 Concepts: development risk by phase, land purchase option	Part 2, sections 2.2 and 2.3
L5 Concepts: FAR, density, efficiency, parking ratio	Part 2, section 2.4
L5 Calculations: net absorption, vacancy rates	Part 2, section 2.8
L6 Concepts: construction lender protections, developer guarantees	Part 3, section 3.2
L6 Concepts: JVs, limited partnerships, preferred equity	Part 3, section 3.5
L6 Concepts: mezzanine debt in the capital stack	Part 3, section 3.4
L6 Concepts: phasing and real options	Part 3, section 3.6
L6 Calculations: construction interest, interest reserve, incremental borrowing cost (incl. mezzanine)	Part 3, sections 3.3 and 3.4
L7 Concepts: abatement vs TIF, tax code favors real estate, taxes at sale, 1031, entities, 2025 act, OZs, LIHTC and credits	Part 4, sections 4.1 to 4.8
L7 Calculations: property tax subsidies, effective and marginal rates, interest and depreciation tax savings	Part 4, sections 4.1 to 4.4
Practice Problems 1, 2 and 3	Part 5

Part 1. Formula sheet

Site and market (Lesson 5)

Measure	Formula	Example
Floor area ratio (FAR)	Gross conditioned area / site area (parking excluded)	684,779 / (2.64 x 43,560) = 5.95
Density	Units / acres	520 / 2.64 = 197.0 DU/AC
Efficiency	Net rentable area / gross conditioned area	575,386 / 684,779 = 84.0%
Parking ratio	Stalls / housing units	518 / 520 = 1.00
Stalls required	Units x zoning ratio	520 x 0.75 = 390
Bid-rent slope	Extra round-trip miles x (1 / speed) x wage x trips a month	2 x 1/20 x $20 x 20 = $40 a month per mile
Income required for a rent	Monthly rent x 12 / 0.30	$1,500 x 12 / 0.30 = $60,000
Affordable rent for an income	Income x 0.30 / 12	$75,000 x 0.30 / 12 = $1,875
Demand funnel	Households x renter % x income-qualified % x capture %	40,000 x .40 x .35 x .05 = 280 units
Vacancy rate	Vacant space / inventory	Vacancy falls 20.0% to 17.0%
Net absorption	Occupied space at end of period minus occupied at start	830,000 - 800,000 = +30,000 SF
Years of supply	(Vacant space + space under construction) / annual net absorption	Industrial 1.4; office 26
Absorption (lease-up) period	Units to lease / units leased per month	520 / 25 = 20.8 months
Break-even rent	(Operating expenses + debt service) / (Units x 12 x occupancy)	($2.1m + $3.9m) / (300 x 12 x 0.885) = $1,883

Financing (Lesson 6, with Lesson 4 terms)

Measure	Formula / rule	Example
EBC (effective borrowing cost)	The IRR of the loan cash flows to the borrower, net of fees: RATE(n, -interest, loan - fee, -loan)	$2m at 6%, $100k fee, 10 yr: 6.70%
IBC (incremental borrowing cost)	Extra interest / extra loan (or RATE on the incremental cash flows if fees or terms differ)	($2,647,938 - $2,344,055) / $2,521,845 = 12.05%
Stacking rule	Add a layer only while its IBC is below the cost of the next source. Stop at the binding constraint	Debt first, then LP equity
LTV	Loan / value	$5.0m / $7.5m = 66.7%
DSCR	NOI / debt service	$450,000 / $300,000 = 1.50x
Max loan, DSCR (interest-only)	NOI / required DSCR / interest rate	$3,188,000 / 1.25 / 6.50% = $39,236,923
Max loan, LTV on value	LTV % x NOI / cap rate	70% x $3,188,000 / 5.50% = $40,574,545
Construction loan size	Loan-to-cost % x total project cost (cost includes fee and interest reserve, so it is circular)	70% x $50,436,905 = $35,305,834
Construction interest	Interest accrues on the opening balance and is added to the loan. Monthly draws at start of month:	FV(7.40%/12, 15, -400,000, 0, 1) = $6,304,691

	FV(r/12, n, -draw, 0, 1)	
Interest reserve	Amount due at payoff - total draws	$6,304,691 - $6,000,000 = $304,691
Combined cost of senior + mezzanine	(Senior % x senior rate + mezz % x mezz rate) / total debt %	(0.65 x 7.15% + 0.05 x 13%) / 0.70 = 7.57%
LP hurdle balance	Opening balance x (1 + pref) - LP tier-1 cash. Split changes when it hits zero	Zero in Year 5 = LP IRR exactly 8%
Tier 1 cash needed to clear hurdle	LP cash to clear hurdle / LP share (90%)	$10,365,408 / 0.90 = $11,517,120
Promote	GP tier-2 cash minus what its capital share would have earned on that cash	$496,576 - $248,288 = $248,288
Real option value	Expected value with the right to stop minus expected value without it	$2.2m - $1.0m = $1.2m

Taxes (Lesson 7)

Measure	Formula	Example
Property tax	Tax rate x assessed value	1% x $10m = $100,000
Cook County tax bill	(Assessed value x equalizer - exemptions) x tax rate. Assessed = market value x 10% (residential) or 25% (commercial)	($5m x 3.03) x 6.85% = $1,037,775
Abatement saving	Tax at full value - tax paid (frozen at pre-improvement value)	$100,000 - $20,000 = $80,000 a year
TIF payment	PV of the property-tax increment (after minus before) over the TIF term, at the city rate	NPV(4.75%, increments) = $65,191,327
Bracket tax	Base tax + marginal rate x (taxable income - threshold)	$206,583.50 + 37% x ($1m - $768,700) = $292,164.50
Effective rate	Total tax / income	$292,164.50 / $1m = 29.22%
Marginal rate	Rate on the next $1. Drives decisions	37%
Interest tax saving	Interest x marginal rate	$36,000 x 37% = $13,320
After-tax interest rate	Rate x (1 - marginal rate)	12% x 0.63 = 7.56%
Annual depreciation	(Cost - land) / tax life. Life is 27.5 yr residential, 39 yr nonresidential	($1.1m - $0.1m) / 39 = $25,641
Depreciation tax saving	Annual depreciation x marginal rate	$36,364 x 37% = $13,455
Adjusted basis	Cost - depreciation taken	$1.1m - $256,410 = $843,590
Depreciation recapture	Lesser of (gain, depreciation taken) x up to 25%	$256,410 x 25% = $64,103
Capital gain tax	(Price - original cost) x 0%, 15% or 20% (plus 3.8% NIIT at high income)	$400,000 x 20% = $80,000
1031 deferral	Recapture tax + capital gains tax deferred. Boot (cash kept, debt relief) is taxed now	$225,000 + $1,800,000 = $2,025,000
New basis after 1031	New price - deferred gain	$15m - $9.9m = $5.1m
Bonus depreciation Year 1	Short-lived property in full + (remaining building basis / life)	$1.1m + $4.4m/27.5 = $1,260,000
LIHTC income limit	AMI x size factor x 60%	$100,000 x 70% x 60% = $42,000
LIHTC rent limit	Income limit x 30% / 12	$42,000 x 0.30 / 12 = $1,050
Blended AMI	Unit-weighted average of each unit's AMI level	(20 x 80% + 20 x 40%) / 40 = 60%

Credit vs deduction	Credit cuts tax dollar for dollar. Deduction cuts tax by deduction x marginal rate	$10,000 credit = $10,000; deduction = $3,700 at 37%

Exam trap

Interest-only means principal is flat. Annual payment = loan x rate. The loan balloons at maturity, so the last-year cash flow includes repaying principal.

Fees cut the loan proceeds, not the loan balance. In EBC, the borrower receives loan minus fee but repays the full loan.

Equal fees cancel in IBC. [L6 s5] RATE(10, -90,000, 1,000,000, -1,000,000) = 9.00% with or without the $100,000 fee.

Part 2. Lesson 5: Site and Market Analysis

2.1 The question every development must answer

Will this market pay the rent this site needs? Break-even rent is what the project must collect to cover costs and the required return. Market rent is what tenants in that location actually pay. The site sets the first. The market sets the second. [L5 s3]

Feasibility test (in order)	What it asks
1. Physical	Soil, depth to bedrock, slope, water
2. Legal	What zoning allows today, and whether it can be changed
3. Design and cost	What fits on the site, how efficiently, and at what cost
4. Financial	Break-even rent against market rent

2.2 The team, developer types and development risk

Development team [s4]

- **Developer:** equity owner and quarterback. Finds the highest and best use and manages risk.
- **Market analyst:** expected rents and absorption (valuation or brokerage background).
- **Financial analyst:** expected returns; structures debt and equity.
- **Architect, engineers, cost estimator:** design the building (form against function).
- **General contractor:** manages construction and subcontractors.

Key idea

Everyone except the developer is paid a fee. **The developer is paid last, from whatever is left**, so the developer carries the residual risk.

Types of developers [s5]: follow the incentive

Type	How paid	Typical risk / failure
Design-led (architect or engineer)	Design first, return second	Too little leasable space for what the building costs
General and limited partnership	Paid with equity. GP is the entrepreneur managing risk; LP is the passive capital	GP earns more when value rises
Fee developer	Cost-plus: a percentage of cost, common on public projects	Earns more when cost rises
Design-build (turnkey)	One firm quotes a fixed price and bears the overruns	Quote is higher because it prices the overrun risk

Approaches and delivery [s6]

- **Site looking for a use:** developer controls the land first and carries market risk. Task is to find the highest and best use. Common in public redevelopment.
- **Use looking for a site:** developer brings a concept or specialty (multifamily, industrial, retail). Often partnered with an anchor tenant or capital source.
- **Build-to-suit (BTS):** tenant commits before construction. It is a use looking for a site: the tenant is the use.
- **Spec:** developer starts with no tenant under contract. Riskier, but some tenants pay for faster delivery.

Four phases: money in, risk down [s7]

Phase	Cumulative investment (% of total cost)	Required return
1. Acquisition (raw land, controlled by option)	5%	40%
2. Planning (entitlement, design, cost and market feasibility)	25%	40%
3. Construction	90%	20%
4. Stabilization and refinance (lease-up, absorption)	100%	10% on the chart; about 8% once stabilized

Key idea

The riskiest phase is the first, when the least money is at stake. Risk falls as money goes in, so required return falls from about 40% to about 8%.

My note: [L5 s7] says "about 8% once stabilized" while its chart shows 10% in the stabilization row. [L6 s3] splits it into lease-up about 10% and stabilized about 8%. If asked, say lease-up about 10%, stabilized about 8%.

2.3 Due diligence and the land purchase option

Physical conditions	Legal and regulatory constraints
Existing structures (demolition cost or preservation objections)	Zoning, and the likelihood of a variance or rezoning
Visibility and access	Easements to utilities and neighbors
Slope, soil, hydrology, depth to bedrock	Covenants and deed restrictions
Environmental: contamination, wildlife, runoff	Impact fees and required exactions
Utilities: water, sewer, power, gas, telecom	

The developer dilemma: finding out whether a site works takes time and money. Once the plan is known, the seller can raise the price or sell to someone else. **A land purchase option solves both.** [s8]

- An option is the **right, not the obligation**, to buy land at a set price by a set date.
- Buyer completes due diligence with no risk of renegotiation. Money at risk stays small until the project is known to be feasible.
- Negotiation: the buyer wants the longest period. **Option price rises with the length of the period and with uncertainty about value.**

Worked example: data center site, Lebanon, Indiana [s9]

Pay $100,000 today for the exclusive right to buy at $10,000,000 within 12 months, while trying to rezone from agricultural to industrial.
Rezoning fails: loss is $100,000 plus legal fees, **not $10,000,000**.
Rezoning succeeds: price is still $10,000,000 on land now worth more.
The option costs 1% of the price ($100,000 / $10,000,000).

2.4 What can the site hold? Four yield measures

Measure	Definition	Chauncey Square	Result
FAR	Gross conditioned area / site area. Parking excluded	684,779 SF / (2.64 ac x 43,560)	5.95

Density	Units / acres	520 / 2.64	197.0 DU/AC
Efficiency	Net rentable / gross conditioned	575,386 / 684,779	84.0%
Parking ratio	Stalls / housing units	518 / 520	1.00
Stalls required	Units x zoning ratio	520 x 0.75	390

- Efficiency is the share of built area that earns rent. The other 16% is corridors, stairs, elevators and amenity.
- Count the garage and only **63.9%** of what is built earns rent: 575,386 / 900,887.

Chauncey Square: what the parking costs [s11]

Line	Calculation	Result
Stalls required by zoning	520 x 0.75	390
Stalls built (two levels below grade)	241 on B1 + 277 on B2	518
Surplus stalls	518 - 390	128
Cost of the surplus	128 x $50,000	$6,400,000
Parking ratio with B1 alone	241 / 520	0.46, under the 0.75 minimum
Cost saved by dropping B2	277 x $50,000	$13,850,000 (not available)

Key idea

A garage comes in **whole levels**. Two levels overbuild by 128 stalls. One level is 149 stalls short (390 - 241) and cannot be permitted. **The parking minimum is the binding constraint**, so the $13.85m saving belongs to a building the city will not approve.

2.5 Where uses locate: bid-rent and agglomeration

Single-user bid-rent [s13]

Every user works out the most it would pay for a location. **The land goes to the highest bid.** Assumptions: all jobs in the CBD reached by one road; wage $20 an hour; travel 20 mph; 20 round trips a month.

- Living one mile farther out: 2 miles x 1/20 hour x $20 = $2 a round trip. $2 x 20 = **$40 a month**.
- Rent falls $40 per mile so nobody gains by moving: **$120 at the center, $0 at 3 miles**.

Competing users [s14]

- Walkers travel 3 mph, so a mile costs them $266.67 a month against $40 for drivers.
- Walkers outbid drivers inside the first mile (bid reaches $346.67 at the center). Drivers hold 1 to 2.75 miles.
- The city ends at 2.75 miles, where the drivers' bid of $10 equals agricultural rent.
- **The steeper curve wins the center.** Land goes to whoever pays the most for being close.

Monocentric and polycentric cities [s15, s16]

- With one center, each use takes the ring where it outbids the rest (Burgess concentric zone model, 1925): business district, transition, working-class homes, residential, commuters. A use needs to pay only $1 more than the next best use.
- **What moves the rings:** a new interchange or transit line changes travel cost; remote work and ride-hailing change trip frequency.
- Who lives closest is not settled by the model. Higher income raises both the value of time and the demand for land. In most US metros lower-income households live nearer the center because they use less land (apartments).

- **Indianapolis is poly-centric.** A monocentric city would show office rent highest Downtown and falling in every direction. Keystone ($24.89) has posted the highest office rent for 18 straight quarters; Downtown is $24.22 with 23.9% vacancy vs 16.9% in North/Carmel and 13.1% in Fishers. Each sub-center has its own bid-rent curve.

Agglomeration economies [s17 to s21]

Concept	Meaning
Agglomeration	Firms and uses gain from locating near one another. A positive spillover (externality)
Urbanization economies	Spillovers from any industry: shared suppliers (lawyers, accountants), shared infrastructure, large general labor pool
Localization economies	Spillovers from the same industry: shared specialized inputs, specialized labor pool (Silicon Valley), knowledge spillovers (West Lafayette)
Why it is a developer tool	A single owner of the whole site collects the spillover in rent. Separate owners cannot: each ignores the customers it sends next door. That is why a shopping center has one landlord
Anchors and inline tenants	Anchors create traffic and pay far less rent per SF. Inline tenants live on that traffic and pay a premium. Mall anchors got a rent subsidy of at least 72% per SF (Pashigian and Gould, 1998)
Mixed-use	A ground-floor grocer, coffee shop or gym can raise apartment rents above. The retail can lose money on its own lease and still raise site value
Developer decision	Rent the best space to the highest bidder, or to the tenant that raises everyone else's rent? A single owner picks the second. Value is the sum across all leases

Retail center types [s20]

Type	Typical size (SF)	Anchors	Anchor share	Trade area
Strip / convenience	Under 30,000	None or a convenience store	Not stated	Under 1 mile
Neighborhood	30,000 to 125,000	1+ (supermarket)	30% to 50%	3 miles
Community	125,000 to 400,000	2+ (discount, supermarket, drug)	40% to 60%	3 to 6 miles
Power center	250,000 to 600,000	3+ category-dominant	70% to 90%	5 to 10 miles
Regional mall	400,000 to 800,000	2+ department stores	50% to 70%	5 to 15 miles
Super-regional mall	Over 800,000	3+ department stores	50% to 70%	5 to 25 miles

Spillover or the same households? [s21]

Qdoba co-tenancy within 0.25 mile (856 locations, observed): Starbucks 52.9% (lift 2.00), Subway 30.7% (1.05), Verizon 30.0% (2.10), McDonald's 24.1% (1.07), Panera 20.2% (3.00), H&R Block 20.0% (1.41). **Lift** compares Qdoba with a typical brand on the same neighbor; 1.0 means no difference.

Key idea

Two stories fit the same data. **Spillover:** each tenant raises the other's sales, so a developer can charge for it. **Same households:** both want the same incomes, traffic and daytime population. **Proximity cannot tell them apart**; co-tenancy is "not a lease clause or proof of causation."

Real spillover evidence: sales fall when the neighbor closes, anchors pay less rent, or leases carry co-tenancy clauses.

In-class decision: which uses belong together? [s22] (my answers; the deck poses these without answers)

1. Classify. A (grocery anchor + coffee shop): **real spillover**. The grocer draws the trip and the coffee shop sells to people already there. B (two competing fast-casual): **same households**, and in a 4-suite ground floor mostly negative since they split one customer pool. C (fitness studio + live-music bar): **negative spillover**. Noise hurts the studio. D (day care + dental office): **same households**, with weak complementarity (parents making one trip).
2. Discount. The grocery anchor gets the rent discount. The other three tenants (inline) pay a premium, so **the inline tenants effectively pay for it** through higher rent, and the owner recoups it through the spillover.
3. Evidence. Pull CREHQ co-tenancy for the coffee brand (how often it sits within 0.25 mile of a grocer). It would not prove causation, only that they cluster.

2.6 The three steps of a market analysis [s12]

A market analysis studies supply and demand for a specific use, to inform a specific decision.

- **1. Scan the environment:** identify the opportunity and the change that creates it; account for public policy and public investment.
- **2. Analyze the local market (demand):** who are the customers today and who will they be?
- **3. Analyze the competition (supply):** existing, pending and planned projects. "There is no low-hanging fruit. Why you, and why has no one done it already?"

Demand is the end users (buyers or renters). Supply is the competition (existing and planned or under construction). **Space that delivers before yours leases first.**

2.7 Demand drivers by property type [s23]

Use	What drives demand	Where to find it
Single-family homes	Households aged 35 to 64, household formation, income and job growth, school quality, mortgage rates	Census ACS, BLS, Freddie Mac
Multifamily rental	Households under 35, renter share, rent relative to income, cost of owning	Census ACS, CoStar
Retail	Households and income in the trade area, traffic counts	Census ACS, state traffic counts, CREHQ
Office	Employment in office-using sectors, space per worker, remote work	BLS, CoStar, broker reports
Industrial	Manufacturing and logistics employment, freight volume	BLS, CoStar, broker reports
Hotel	Air passengers, conventions, tourism	Airport and visitor bureau data

Indianapolis in five numbers (ACS 2024) [s24, s25]

Measure	Value	Implication
Population	2,173,288	Size of the market for every use
Households	869,616, avg size 2.46	Housing demand is counted in households, not people
Age	Median 37.4; ages 25 to 34 are 14.4%	Prime renting ages are one resident in seven
Household income	Median $80,239	Sets what a household can pay in rent

		and spend
Tenure	66.7% own, 33.3% rent	Rental pool is one household in three

- Median gross rent $1,273; 48.1% of renter households pay 30%+ of income in rent.
- Renter share by age: 15 to 24 = 84.1%, 25 to 34 = 57.8%, 35 to 44 = 30.6%, then about 22%.
- Households under 35 are 23% of all households and 43% of renters. **Growth in ages 25 to 34 drives apartment demand; growth in ages 35 to 54 drives home buying.**

Calculation: income required for a rent [s26]

Worked

Income required = monthly rent x 12 / 0.30. $1,500 x 12 / 0.30 = **$60,000**. (30% of gross income is the HUD cost-burden line.)
Your turn: $75,000 x 0.30 / 12 = **$1,875** a month. 29.0% of renter households (83,912) earn $75,000+, versus 65.4% of owners and 53.3% of all households. A building asking $1,875 is priced for 29% of today's renters. Median renter pays $1,273.

Calculation: the demand funnel [s27]

Step	Assumption	Households
Households in the trade area		40,000
Renter households	x 40%	16,000
Income-qualified at the asking rent	x 35%	5,600
Captured by this project	x 5%	280

- Supportable units = 40,000 x 0.40 x 0.35 x 0.05 = **280**. The Chauncey Square study fits 520. **If the market supports 280, the market is the binding constraint, not zoning.**
- **Trade area:** where most customers come from. Draw it by drive time, not radius (a river, rail line or missing bridge cuts a 3-mile circle). Every funnel number changes when the boundary moves.
- The capture rate is the weakest assumption.

In-class exercise: two trade areas, one 300-unit project at $1,875 [s28]

1. Funnel. North: 52,000 x 22% = 11,440 renters; x 55% = 6,292 qualified; x 4% = **251.7 units**. Downtown: 28,000 x 62% = 17,360; x 30% = 5,208; x 6% = **312.5 units**.
2. Which supports 300 units? Downtown (312.5 > 300). North (251.7) does not.
3. Qualified renters per new unit (competition plus yours). North: 6,292 / (200 + 300) = **12.6**. Downtown: 5,208 / (1,200 + 300) = **3.5**. Now **North** is better, because Downtown has 1,200 competing units coming.
4. Both at 5%. North: 6,292 x 5% = **314.6**. Downtown: 5,208 x 5% = **260.4**. The answer flips: North supports 300, Downtown does not. **The conclusion hinges on the capture rate, the weakest assumption**, so stress it and weigh the competing pipeline.

2.8 Supply: what a broker report measures

Term	Meaning	Watch for
Inventory	Total existing space the firm tracks	Firms track different building sets
Vacancy rate	Vacant space / inventory	Vacant means ready to occupy now
Availability rate	Space being marketed / inventory	Includes space a tenant has not yet left
Net absorption	Change in occupied space over the period	Not the same as leasing activity

Under construction	Space that will deliver soon	Speculative or build-to-suit
Asking rent	Rent quoted on available space	Excludes free rent and tenant improvements
Cap rate	NOI / sale price on recent sales	Rarely in a leasing report; use CoStar

Rents are not quoted on one basis: Indianapolis office is full-service gross, retail and industrial are triple-net, apartments are per unit per month. [s29]

Net absorption and vacancy [s30]

Worked

Net absorption = occupied space at end of period - occupied space at start.
A market has 1,000,000 SF. Occupied goes from 800,000 to 830,000 SF. Net absorption = **+30,000 SF** (3.0% of inventory). Vacancy goes from 20.0% (200,000 / 1,000,000) to **17.0%** (170,000 / 1,000,000).
Positive: tenants occupy more space than before. Negative: they occupy less.

Exam trap

Leasing activity is a different series. A tenant that signs 50,000 SF and vacates 50,000 SF elsewhere adds 50,000 SF of leasing and **zero** absorption.
Vacancy can rise in a year of positive absorption if more space is delivered than absorbed.
Indianapolis Q2 2026: retail leasing activity was 225,000 SF, yet net absorption was -40,190 SF. Space was leased and occupied space still fell.

Indianapolis by property type, Q2 2026 [s31]

	Office	Industrial	Retail	Multifamily
Publisher	CBRE	CBRE	Cushman & Wakefield	Kirkland
Inventory	35.99M SF	346.62M SF	44.93M SF	195,558 units
Vacancy	19.9%	5.9%	4.9%	11.5%
Net absorption (qtr)	+250,748 SF	+3.06M SF	-40,190 SF	Not reported
Under construction	48,012 SF	6.58M SF	62,661 SF	4,732 units
Asking rent	$22.37 FSG	$6.43 NNN	$17.29 NNN	$1,342 / unit / mo

Developers build where vacancy is low: 6.58M SF of industrial under construction vs 48,012 SF of office.

Two brokers, one quarter [s32]

Office Q2 2026, CBRE vs Cushman & Wakefield: inventory 35.99M vs 40.31M SF; vacancy 19.9% vs 20.0%; quarterly net absorption +250,748 vs +28,828 SF (nearly 9x apart); under construction 48,012 vs 97,159 SF. **Each firm tracks a different set of buildings**, and over six months the two are closer than over three. A broker report is marketing by a firm paid on leases and sales. **Read two and cite the one you use.**

Calculation: years of supply [s33]

Years of supply = (vacant space + space under construction) / annual net absorption

Industrial: vacant 5.9% x 346.62M = 20.45M SF; plus 6.58M under construction = 27.03M; / 19.0M SF annual absorption = **1.4 years**.
Office: vacant 19.9% x 35.99M = 7.16M SF; plus 0.05M = 7.21M; / 0.28M (first-half absorption x 2) = **about 26 years**.
Industrial clears in under 18 months, which is why developers are building spec again. Office would take a generation.

Where the data come from [s34, s35 to s37]

Source	Gives you	Paid by
Census ACS (data.census.gov)	Population, households, age, income, tenure, rent. Tables DP02 (households), DP03 (income), DP04 (tenure, rent), DP05 (age), B25007, B25118	Taxpayers
BLS and FRED	Employment by sector, unemployment, wages	Taxpayers
HUD	Area median income and income limits	Taxpayers
Broker reports (CBRE, C&W, Kirkland)	Vacancy, absorption, pipeline, asking rent by submarket	Firms paid on leases and sales
CoStar	Rents, vacancy, absorption, sales and cap rates by property	Subscribers
CREHQ	Tenant site criteria, locations, co-tenancy	Subscribers

- **Public data are slower and neutral. Private data are faster and have a seller.** Cite publisher and date for every figure.
- **CREHQ reports:** where a brand operates, published site criteria (size, format, traffic, population, income), observed patterns around existing stores, co-tenancy, franchise costs and planned openings, Site Selector.
- **CREHQ does not report:** rents, vacancy and absorption, construction pipeline, cap rates and sale prices, demographics for a site you choose, foot traffic, or whether a brand wants your site.
- **Published vs observed:** published is what the brand says it wants; observed is where it operates. Domino's publishes a 20,000-vehicle minimum, yet half its stores sit on roads under 13,630. **A published number is a preference, not a rule.** A blank is unknown, not zero or a pass.
- **Binding criterion:** the criterion that fails first. For Handel's Ice Cream on the Indiana endcap example, income is far below its stores, and the brand never published an income minimum. A fit is a research lead, not a leasing commitment.

2.9 Absorption, lease-up and break-even rent

Absorption period [s38]

Units leased per month	Months to lease 520 units
15	34.7
25	20.8
40	13.0

Absorption period = units to lease / units leased per month. **Every month of lease-up is a month of construction-loan interest with little rent to pay it**, so slower lease-up needs more equity or a bigger interest reserve. The absorption rate comes from how fast comparable new buildings filled. Lease-up period is completion to stabilized occupancy.

Break-even rent against market rent [s39]

Occupancy the market supports	Occupied unit-months	Break-even rent / unit / month
95.0%	3,420	$1,754
88.5% (Indianapolis average)	3,186	$1,883

80.0%	2,880	$2,083

Break-even rent = (operating expenses + debt service) / (units x 12 x occupancy). Project: 300 units, $2,100,000 expenses, $3,900,000 debt service. **The lower the occupancy the market supports, the higher the rent the project needs.** Market average is $1,342 for 50+ unit properties, but new buildings rent above average, so compare against new comparables in the submarket. **If break-even is above what comparable new buildings get, the project fails the financial test: cut cost, add density, or walk.**

Lesson 5 review [s40]

- The developer is paid last and decides first. A land option buys time for due diligence.
- FAR, density, efficiency and parking ratio measure what a site holds.
- Uses bid for land and the highest bid takes it. One center gives rings; several give sub-centers. A single owner captures the spillover between tenants.
- Count households, then renters, then those who qualify. Income required is rent x 12 / 0.30. Capture rate is the weakest assumption.
- Net absorption is the change in occupied space. Read two brokers and name the publisher. Compare break-even rent with market rent.

Part 3. Lesson 6: Financing Development

Running example: 200 apartment units in Indianapolis. All figures are teaching assumptions. SOFR assumed at 3.90%.

3.1 Who pays for a building before it earns rent?

Phase	Required return	Who funds it	What they hold
Acquisition and planning	~40%	Developer's own equity and a land option	The option, plans, entitlements
Construction	~20%	LP equity first, then the construction loan	Mortgage on a half-built building and developer guarantees
Lease-up	~10%	Construction loan and its interest reserve	A finished building with few tenants
Stabilized	~8%	Permanent loan and the equity that remains	A building with rent

Key idea

Each phase is funded by the cheapest money willing to take its risk.

The capital stack [s4]

Layer (top to bottom)	Share of cost	Dollars	Position
GP equity	3%	$1.51m	First loss, paid last, earns the promote
LP equity	27%	$13.62m	Paid after all debt. Preferred return, then a share
Mezzanine debt	not used		Secured by the ownership interests, not the building
Senior construction loan	70%	$35.31m	Mortgage on the property. Paid first. Cheapest

Top loses first and earns most. Bottom is paid first and costs least. Each layer is added only while it is the cheapest money left, and only as far as its lender or investor allows. At the end of Year 3 a $39.24m permanent loan replaces the construction loan.

Sources and uses [s6]

Uses	Amount	Sources	Amount
Land	$3,000,000	Construction loan (70% of cost)	$35,305,834
Hard cost ($180,000/unit)	$36,000,000	LP equity (90% of equity)	$13,617,964
Soft cost (20% of hard)	$7,200,000	GP equity (10% of equity)	$1,513,107
Construction loan fee (1%)	$353,058		
Interest reserve	$3,883,847		
Total project cost	**$50,436,905**	**Total sources**	**$50,436,905**

$252,185 per unit all-in. Build 18 months, lease up 12 months at about 17 units a month. Lender requires six months of stabilized operations, so the permanent loan funds at the end of Year 3.

3.2 Construction loans, lender protections and guarantees

Construction loan vs permanent loan [s7]

	Construction loan	Permanent loan
Job	Pays for the building until leased	Repays the construction loan
Sized on	70% of total project cost	Lesser of 1.25x coverage and 70% of value at a 5.50% cap
This project	$35,305,834	$39,236,923
Rate	7.50% floating (SOFR + 360 bp)	6.50% fixed
Interest	Accrues; paid from the interest reserve	Paid from NOI every year, interest-only
Funded	In draws, after the equity	All at once, end of Year 3
Term and fee	3 years, 1% fee	10 years, 1% fee
Recourse	Developer guarantees	None, except carve-outs

Key idea

The construction lender sizes on cost because there is no income yet. The permanent lender sizes on income.

How construction lenders protect themselves [s8]

- **Loan-to-cost cap** (70% here): equity carries the first 30%.
- **Equity first:** every equity dollar is spent before the first loan dollar.
- **In balance:** if the undrawn loan cannot finish the job, the developer deposits the difference.
- **A take-out the lender believes in:** it sizes the permanent loan before it lends.
- **Draws against inspections:** an engineer certifies the work before each draw.
- **Retainage and lien waivers:** 5% to 10% held back; each draw paid against releases.

Who pays for an overrun? Not a mezzanine lender (mezzanine is arranged at closing, if at all). **The developer does**: the completion guarantee and in-balance clause require new equity. That is why overrun risk sits with the GP, not the LP.

The developer's guarantees [s9]

Guarantee	Covers	This project	Burns off
Completion	Finish the building and pay every overrun	10% overrun on hard + soft cost: $4,320,000	At completion
Carry	Interest, taxes and operating deficits until stabilized	Six months of delay: $1,323,969 of interest	At stabilization
Repayment (25%)	A quarter of the loan if not repaid	$8,826,458	At the take-out
Carve-outs	Losses from fraud, waste or voluntary bankruptcy	Unlimited if triggered	Never; permanent loan has them too

Key idea

The GP has $1,513,107 of equity but guarantees $8,826,458 of the loan: **5.8 times its equity.** The guarantees, not the equity, are the developer's largest exposure. Phasing caps them.

3.3 Construction interest and the interest reserve

Annual draws [s10]

	Year 0	Year 1	Year 2	Year 3	Total
Costs paid	$5,513,058	$28,560,000	$12,480,000	$0	$46,553,058
Equity	$5,513,058	$9,618,013	$0	$0	$15,131,072
Loan for costs	$0	$18,941,987	$12,480,000	$0	$31,421,987
Interest added	$0	$0	$1,420,649	$2,463,198	$3,883,847
Loan balance	$0	$18,941,987	$32,842,636	$35,305,834	

- **Equity goes first.** Interest is 7.50% of the opening balance each year and is added to the loan: that is the **interest reserve**.
- Year 2 check: 7.50% x $18,941,987 = $1,420,649. Year 3: 7.50% x $32,842,636 = $2,463,198.
- **Circular:** the reserve is a cost, the loan is 70% of cost, and interest depends on the loan. Iterate until they agree.

What the construction loan really costs [s11]

- Lender cash flows: Year 0 receives the $353,058 fee; Years 1 and 2 advances $18,941,987 and $12,480,000; Year 3 repaid $35,305,834. **IRR = 8.34% a year against a 7.50% rate.**
- The 1% fee is charged on the full commitment but the money is out less than two years on average. The borrower's EBC is the same 8.34%.
- **Assignment shortcut** (draw everything at Year 0): three years of interest on $35,305,834 at 7.50% = $8,554,493, not $3,883,847. It overstates interest, so treat it as an **upper bound**.

Practice: construction loan interest [s14, s15]

Draws of $400,000 at the **beginning** of each month for 15 months. Rate = SOFR 3.90% + 350 bp = 7.40%, accrued monthly, nothing paid until month 15. Fee 1% of amount due at payoff.
1. Amount due, month 15 (before fee): FV(7.40%/12, 15, -400,000, 0, **1**) = **$6,304,691**. (Monthly rate 0.6167%; type 1 because draws come at the start of the month.)
2. Interest reserve: $6,304,691 - (15 x $400,000 = $6,000,000) = **$304,691**.
3. Payoff with fee: $6,304,691 x 1.01 = **$6,367,738**.
4. Yield vs 7.40%: above, 8.88% a year (nominal). The fee is paid on money outstanding only 8 months on average, so the fee adds yield.
Note: the reserve is part of the loan, so a real lender sizes the loan to include it.

3.4 IBC, the stacking rule and mezzanine debt

IBC: the rule that builds the stack [s5]

	Lesson 4, Example 5	Same idea, with fees
Smaller loan	$4,000,000 at 5%	$2,000,000 at 6%, $100,000 fee
Larger loan	$5,000,000 at 6%	$3,000,000 at 7%, $100,000 fee
Term	10 years, interest-only	10 years, interest-only
EBC of each loan	5.00% and 6.00%	6.70% and 7.49%
Extra interest on extra $1,000,000	$100,000 a year	$90,000 a year
IBC of the extra $1,000,000	10.00%	9.00%

Key idea

Stacking rule: add a layer only while its IBC is below the cost of the next source. Stop at the binding constraint (loan-to-cost, coverage at the take-out, lender consent, the LP's terms).
Price ranks the layers. The binding constraint decides where each one stops.

IBC 1: stretch the construction loan to 70% [s12]

Same $50.44m budget	Base quote	Stretch
Share of total project cost	65%	70%
Rate on whole balance	7.15% (SOFR + 325)	7.50% (SOFR + 360)
Loan	$32,783,988	$35,305,834
Interest a year, fully drawn	$2,344,055	$2,647,938

IBC 1 = ($2,647,938 - $2,344,055) / ($35,305,834 - $32,783,988) = $303,882 / $2,521,845 = **12.05%**. The next source is LP equity at about 15% (IBC 4), so 12.05% is cheaper: **take the stretch**. The binding constraint is the lender stopping at 70% of cost. One more test: can the take-out repay 70%?

IBC 2: the take-out and negative leverage [s13]

Permanent loan test	Calculation	Max loan
Coverage 1.25x	$3,188,000 / 1.25 / 6.50%	$39,236,923
Value: 70% at 5.50% cap	70% x $3,188,000 / 5.50%	$40,574,545
The loan: the smaller	Coverage binds	$39,236,923

- **Why borrow at 6.50% against a 5.50% cap?** Cash yield falls (5.67% unlevered, 3.84% levered in Year 4), but IRR rises (9.14% unlevered, 13.19% levered) because value grows. **Negative leverage on cash, positive by the Lesson 4 rule.**
- Take-out: $39,236,923 less the 1% fee repays $35,305,834 and returns **$3,538,720** to equity. The loan costs 6.64% with fee; the next source, LP equity, costs about 15%.
- Stabilized NOI $3,188,000; value $57,963,636 at a 5.50% cap.

IBC 3: mezzanine or a bigger senior loan? [s16]

- **Mezzanine debt** sits between senior debt and equity. It is **secured by the ownership interests, not the building**, so the senior lender still faces a new owner if mezz forecloses. That is why mezzanine is available even when the senior lender prohibits second mortgages.
- Combined cost of 65% senior at 7.15% + 5% mezzanine = (0.65 x 7.15% + 0.05 x mezz rate) / 0.70.
- Mezz quoted at 13%: combined **7.57%** vs **7.50%** for the stretch. So the stretch wins.
- **Break-even mezzanine rate = 12.05%**, the IBC of the stretch. Pay less than that and mezz wins; more and it loses.
- Above 70% the senior lender forbids more debt: that is the binding constraint.

Practice: mezzanine with equity capped [s17, s18]

You have $3m of equity and no more. $15m office building, 16% unlevered IRR over 3 years, opportunity cost of equity 15%. Primary lender prohibits second mortgages. Option A: $12m at 11%. Option B: $10m at 9%, so you need $2m of mezzanine.
1. IBC of the extra $2m in A: extra interest = $12m x 11% - $10m x 9% = $1,320,000 - $900,000 = $420,000. IBC = RATE(3, -420,000, 2,000,000, -2,000,000) = **21.00%**.
2. The most you would pay for $2m of mezz: 21%, the IBC of the extra $2m in Option A.
3. Mezz quoted at 14%: B + mezz total interest = $900,000 + $280,000 = $1,180,000 on $12m: RATE(3, -1,180,000, 12,000,000, -12,000,000) = **9.83%** vs **11.00%** for A. **Choose B.**
Why 21% and not 15%? Equity is capped, so the alternative to mezz is the bigger senior loan, not equity. With

equity available at 15%, the most is 15%. Borrowing at all is positive leverage: 16% unlevered IRR exceeds both options.

3.5 Partnerships, joint ventures and preferred equity

Why real estate uses partnerships [s19]

Entity	Owners' liability	Tax	Can it pay a promote?
C corporation	Limited	Twice: company, then shareholder	Not without a second tax on every dollar
S corporation	Limited	Once	No: one class of stock, payouts pro rata
General partnership	Unlimited for every partner	Once	Yes
Limited partnership	GP unlimited; LPs limited to capital	Once; losses flow through	Yes
LLC	Limited for every member	Once (taxed as partnership by default)	Yes

Four reasons: **limited liability for investors, one level of tax, construction losses flow through, and a promote.** An LLC is a company, not a corporation.

Joint venture and single-purpose entity [s20]

- A **joint venture** is a partnership for a single project. It is not an entity separate from its members, so members can face unlimited liability. So venturers hold interests through LLCs.
- A **single-purpose entity (SPE)** owns one project and nothing else. The lender underwrites one building and its debts stay inside the entity. The LP sees one project's books and one waterfall. Phased projects use one entity and one loan per phase.
- Running example: owner is one LLC formed for the building; GP is the developer's LLC (manager); LP is a member with no management role; lender holds a mortgage and developer guarantees.

GP vs LP: who brings what [s21]

	General partner	Limited partner
Capital	10%: $1,513,107	90%: $13,617,964
Brings	Site, plan, team, guarantees	Money
Control	Runs the project day to day	Votes only on major decisions: sale, refinance, removing the GP
Liability	Unlimited as GP (so usually held through an LLC)	Limited to its capital
Fees	Development fee, asset management fee	Pays them
Guarantees to lender	Completion, carry, repayment	None
Share of cash	10% until LP earns 8%, then 20%	90% until it earns 8%, then 80%

Preferred return, true hurdle, promote [s22]

Term	Meaning	In the example
Pari passu	Side by side, in proportion to capital	90/10 until the LP earns 8%
Preferred return	A return the LP receives before the GP's extra share begins	8% a year, compounding

True hurdle	An IRR test: a hurdle balance grows at 8% and falls with each LP distribution. The split changes when it reaches zero	Clears in Year 5
Promote	The GP's share above its share of the capital	20% received vs 10% contributed: 10 points of tier-2 cash
Catch-up	A tier giving the GP extra cash until its share of profit reaches the promote percentage	Not used here

Exam trap

Promote always means the GP's extra share. The LP's 80% is never a promote.

Preferred equity vs LP equity [s23]

	Preferred equity	LP equity with a preferred return
Return	Fixed, e.g. 10%	8% preferred, then 80% of everything above
Upside	None beyond the fixed return	Yes, above the hurdle
Paid	Before the GP's equity, until redeemed	Side by side with the GP until the hurdle
Term	A redemption date	Until the project is sold
If unpaid	Can take control of the project	Votes on major decisions
Loses first	The GP's equity	Both, in proportion
Behaves like	Expensive debt	Equity

Same name, different claim: preferred equity is a fixed claim with a deadline; an LP's preferred return is a priority inside a shared upside.

Who bears the risk? Three structures [s24, s27]

Each cell is the IRR when the Year 5 sale price is weak / base / strong (weak = 6.34% exit cap, base = 5.50%, strong = 5.04%).

Structure	Money partner: weak / base / strong	GP: weak / base / strong
Pari passu 90/10	8.0% / 15.7% / 20.0%	8.0% / 15.7% / 20.0%
8% hurdle, then 80/20	8.0% / 15.0% / 18.9%	8.0% / 21.7% / 28.3%
Preferred equity at 10%	10.0% / 10.0% / 10.0%	-18.6% / 42.1% / 57.1%

- **IBC 4:** the cost of LP equity is what the LP expects: **15.0% in the base case**. It is an expected cost, not a coupon: low when the project disappoints, high when it succeeds. That is the insurance the GP buys.
- Preferred equity shifts all the risk and upside to the GP (it keeps everything above 10%, and can lose all of it). Hurdle-and-promote shares the downside with the LP.
- **Equity is the most expensive layer, so debt goes in first, up to its binding constraint.**

Running example waterfall [s25, s26]

LP hurdle balance	Year 1	Year 2	Year 3	Year 4	Year 5
Opening	$13,617,964	$14,707,402	$15,883,994	$12,535,265	$12,878,170
Plus 8%	$1,089,437	$1,176,592	$1,270,719	$1,002,821	$1,030,254
Less LP tier-1 cash	$0	$0	($4,619,448)	($659,916)	($13,908,424)

Closing	$14,707,402	$15,883,994	$12,535,265	$12,878,170	$0

	Year 0	Year 3	Year 4	Year 5
Project cash to equity	($15,131,072)	$5,132,720	$733,240	$23,666,488
Tier 1, 90/10		$5,132,720	$733,240	$15,453,804
Tier 2, 80/20		$0	$0	$8,212,684
LP	($13,617,964)	$4,619,448	$659,916	$20,478,571
GP	($1,513,107)	$513,272	$73,324	$3,187,917

- **Project IRR 15.71%, LP 14.96%, GP 21.69%.** LP NPV at 12% = $1,709,547. Multiples: LP 1.89x, GP 2.49x.
- Year 3 cash = take-out $3,538,720 + six months of rent $1,594,000. In Year 5, tier 1 needs $15,453,804 of total cash: $13,908,424 to the LP and $1,545,380 to the GP.
- **Promote:** GP receives $1,642,537 of tier 2 vs $821,268 at its 10% share. The extra $821,268 is the promote.

Practice: a three-year waterfall [s28, s29] (re-computed)

Equity $10m: LP $9m, GP $1m. Cash to equity: Y1 $0; Y2 $1,000,000; Y3 $14,000,000 (incl. sale). Terms: 90/10 until LP earns 8% IRR, then 80/20.

1. LP hurdle balance. Y1: $9,000,000 x 1.08 = $9,720,000. Y2: x 1.08 = $10,497,600, less LP tier-1 cash $900,000 (90% of $1m) = $9,597,600. Y3: x 1.08 = **$10,365,408**, less LP tier-1 cash $10,365,408 = $0.

2. Year 3 tiers. Tier-1 total = $10,365,408 / 0.90 = **$11,517,120**. Tier 2 = $14,000,000 - $11,517,120 = **$2,482,880**.

3. IRRs. LP cash Y3 = $10,365,408 + 80% x $2,482,880 = $12,351,712. GP = $1,151,712 + $496,576 = $1,648,288. **LP IRR 14.13%. GP IRR 20.95%. Project 14.85%.**

4. Promote: GP gets 20% x $2,482,880 = $496,576 vs 10% x $2,482,880 = $248,288 at its capital share. **Promote = $248,288.**

3.6 Phasing and real options

Why phase? To cap the guarantees [s30]

Developer exposure	One phase of 200 units	Both phases at once
GP equity at risk	$1,513,107	About twice as much
Repayment guarantee (25%)	$8,826,458	About twice as much
Completion (10% overrun)	$4,320,000	About twice as much
Carry (6 months delay)	$1,323,969	About twice as much

- Phasing caps the guarantees one phase at a time: phase 1's burn off at its take-out, before phase 2 signs.
- Each phase climbs down the Lesson 5 ladder from ~40% to ~8% before the next one starts.
- **Phasing moves risk; it does not remove it. Someone else bears it, and charges for it.**

Phasing tools and who takes the risk [s31]

Tool	Who takes the risk (and charges for it)
Rolling land options and takedowns (buy each parcel when its phase starts)	Land seller: holds an option and charges a premium
One SPE and one loan per phase; guarantees capped and released phase by phase	Lender: underwrites one phase at a time
Pre-leasing or absorption threshold before the next phase starts	Developer (waits)
Pad sales (sell sites to users who build their own); ground lease (control land without buying)	Land buyers / landowner

Public partners: TIF and public infrastructure for shared costs	City: carries infrastructure through TIF
Fund one phase, not a program	LP

What phasing costs [s32]

- Lost economies of scale: one mobilization, one bid, one crane.
- Cost growth: 3% a year on $36.0m of hard cost is **$3,338,172** over three years.
- Building next to an occupied phase 1: noise, staging, safety.
- A market that turns before the later phases. Option premiums on land that may never be used.
- **Cross-collateral trap:** pledging a finished phase 1 to borrow for phase 2 ties phase 2's risk back to phase 1, the opposite of what phasing is for.

IBC 5: funding phase 2's equity [s33]

Phase 1 at end of Year 3	Take-out loan	Cash-out refinance
Lender's minimum coverage	1.25x	1.10x
Rate, interest-only	6.50%	7.00%
Binding test	Coverage: $39,236,923	70% of value: $40,574,545
Interest a year	$2,550,400	$2,840,218
Coverage at that loan	1.25x	1.12x

IBC 5 = ($2,840,218 - $2,550,400) / ($40,574,545 - $39,236,923) = $289,818 / $1,337,622 = **21.7%** (21.9% with the 1% fees). New LP equity for phase 2 expects about 15%. The cash-out costs 21.7%, raises only $1,337,622, and cuts phase 1 coverage to 1.12x. **Raise new LP equity. Keep phase 1 out of phase 2's loan.**

The value of the right to stop: real options [s34]

Worked: expected value with and without the option (50/50 odds; values are present values)

Build both phases now: half of +$8.0m (strong) and half of -$6.0m (weak) = **+$1.0m**.
Wait for phase 1 to stabilize, then decide: strong build +$4.7m, weak stop -$0.3m (the land option premium). Half of each = **+$2.2m**.
The right to stop is worth $1.2m ($2.2m - $1.0m), even though waiting costs $3.3m of cost growth in a strong market. Probabilities and values are assumptions.

Key idea

A **real option** is the right, not the obligation, to expand, wait or abandon. Like the land option: pay a premium now for the right, not the obligation.
Blue Cross Blue Shield Tower, Chicago [s35, s36]: built in 1997 with foundations, columns, riser space and elevator bays sized for 57 stories (the premium, never published). In 2006 the company outgrew 33 stories and chose to build up (the trigger). 2007 to 2010: 24 stories and 920,000 SF added above an occupied building, contract $210m (exercise). The alternative, a second tower, would have cost about $15m more. Capacity doubled to almost 8,000 employees.

In-class exercise: phase 2's capital stack [s37, s38] (re-computed)

Total cost $55,000,000; stabilized NOI $3,300,000; construction quotes 65% at 7.15% or 70% at 7.50% on the whole balance; permanent loan 6.50% IO, lesser of 1.25x and 70% of value at 5.50% cap, 1% fee; phase 2 LP is 90/10, 8% hurdle then 80/20, and invests only if the construction loan is at most 65% of cost.
1. Loans and IBC. 65% = **$35,750,000**; 70% = **$38,500,000**. Interest: $38.5m x 7.50% = $2,887,500 vs $35.75m x 7.15% = $2,556,125; extra $331,375 / extra $2,750,000 = **12.05%**.

2. Permanent loan. Coverage: $3,300,000 / 1.25 / 6.50% = **$40,615,385**. Value: 70% x $3,300,000 / 5.50% = $42,000,000. Loan = $40,615,385; **$40,209,231 after the 1% fee, which repays either construction loan**.
3. Which constraint binds? Price favors 70% (12.05% beats LP equity at ~15%) but **the LP's 65% limit binds**. Senior $35,750,000; equity $19,250,000: **LP $17,325,000 (90%), GP $1,925,000 (10%)**.
4. Cash-out of phase 1? No. About 22% (IBC 5), raises a fraction of the equity needed and ties phase 1 to phase 2.

Lesson 6 review [s39]

- Construction loans are sized on cost; equity goes first. Interest accrues to a reserve, so loan and reserve are circular. The guarantees are the developer's largest exposure.
- Add a layer while its IBC is below the next source; stop at the binding constraint. The take-out sizes on income ($39.24m here).
- Partnerships: one level of tax, limited liability, a promote. A true hurdle is an IRR test. Promote is the GP's share above its capital share.
- Phasing caps guarantees one phase at a time, moves risk and someone charges for it. The right to stop is a real option.

Part 4. Lesson 7: Taxation of Real Estate

4.1 Property taxes: abatement vs TIF

Three levels of tax: **local** (property tax on value, due every year), **state** (income tax; flat in Indiana and Illinois), **federal** (income tax at rising rates, plus tax on gains at sale). The code favors real estate through abatements and TIF, interest and depreciation deductions, lower and deferred capital gains tax, and credits for historic and affordable housing. [s3]

- **Property tax = rate x assessed value** of land and building. Due every year whether or not the property earns income.
- Indiana assesses at market value and caps taxes: 1% homes, 2% apartments, 3% commercial. Chicago: 2.07% apartments, 5.19% commercial, on market value.
- **Effective property tax rate = tax / market value.**

Abatement vs TIF [s5]

$2,000,000 building + $8,000,000 of improvements, taxed at 1%.

	Abatement	TIF
Tax before improvement	$20,000	$20,000
Tax the owner pays after	Capped at pre-improved value: $20,000 a year for 10 years	Full improved value: $100,000 a year
The $80,000 a year increase (the increment)	Never paid; owner keeps it as lower taxes	Paid; developer gets its present value up front as a lump sum
Paid for by	Taxes the city gives up on this property	This property's tax increase, and possibly its neighbors' increases
Developer's benefit arrives	Slowly, as lower taxes each year	At the start, cutting what must be financed

- After 10 years the abated property pays the full $100,000.
- **The city borrows against the future increment to fund the TIF lump sum.**

How a TIF payment is determined [s8] (Assignment 5)

- Site bought for $10,000,000; tax rate 1.90%; 3% growth.
- **1. Tax before improvement** (land): $10,000,000 x 1.90% = $190,000 in Year 1, growing 3% to $201,571 in Year 3.
- **2. Tax after improvement** (from Year 3): Year 3 NOI before property taxes $14,020,506 / 6.50% terminal cap = $215,700,092 market value; x 1.90% = $4,098,302.
- **3. Increment = after - before** = $4,098,302 - $201,571 = $3,896,731 in Year 3; $0 in Years 1 and 2.
- **4. TIF payment = PV of the increment, Years 3 to 25, at the city's 4.75% rate = $65,191,327**, paid as a lump sum at Year 0. It cuts cost before the construction loan is sized. Excel: NPV(4.75%, increments Years 1 to 25).
- Group project: leased County land pays no tax before improvement, so the whole tax is increment; Appendix E discounts at 4.0%.

Chicago property taxes [s6, s7]

- Illinois standard is 33 1/3% of market value. Cook County assesses less, by use: **residential (incl. apartments) 10%, commercial 25%.** State equalizer 3.03 for 2025 lifts Cook to about one third of value. Appendix E 35% rule: commercial up to 35% of SF means treat all as residential.

- **Tax bill = (assessed value x equalizer - exemptions) x local tax rate.** Effective rate on market value = level x equalizer x tax rate: residential 10% x 3.03 x 6.85% = 2.0756% (RFP 2.07%); commercial 25% x 3.03 x 6.85% = 5.1889% (RFP 5.19%).

Worked: Harrison Square parcel (200,000 rentable SF, 10,000 SF retail)

1. Year 3 NOI before property taxes, after ground rent = $3,000,000.
2. Market value = $3,000,000 / 6.00% terminal cap = **$50,000,000**.
3. Retail = 10,000 / 200,000 = 5% of SF, under 35%: **all residential**.
4. Assessed value = 10% x $50,000,000 = **$5,000,000**. 5. Equalized = $5,000,000 x 3.03 = **$15,150,000**.
6. Tax = ($15,150,000 - $0) x 6.85% = **$1,037,775**. RFP shortcut: $50m x 2.07% = $1,035,000 ($2,775 lower).

4.2 Federal income tax: brackets, effective and marginal rates

Midterm sheet: **do not memorize brackets.** Know the mechanics. 2026 brackets are on the slide [s9]; the formula is **Tax = base tax + rate x (taxable income - threshold)**, where base tax is the tax on all income below the threshold. Taxable income = income - deductions (standard deduction $16,100 single, $32,200 joint).

Rate	Single over	Base tax	Joint over	Base tax
10%	$0	$0	$0	$0
12%	$12,400	$1,240	$24,800	$2,480
22%	$50,400	$5,800	$100,800	$11,600
24%	$105,700	$17,966	$211,400	$35,932
32%	$201,775	$41,024	$403,550	$82,048
35%	$256,225	$58,448	$512,450	$116,896
37%	$640,600	$192,979.25	$768,700	$206,583.50

- Most real estate income is taxed to individuals: partnership and LLC income passes through to owners; REIT dividends are taxed to shareholders.
- **Effective rate = total tax / total income** (the average rate). **Marginal rate = rate on the next $1** (the rate for decisions: what a deduction saves, what more income costs).

Worked: single filer [s11]

Earns $10,000: taxable income = $10,000 - $16,100 = $0 (never below $0). Tax $0; effective 0%; marginal 0%.
Earns $60,000: taxable $43,900. 10% x $12,400 = $1,240; 12% x ($43,900 - $12,400) = $3,780. **Tax $5,020. Effective 8.37%. Marginal 12%.**

Practice: married couple, $1,000,000 taxable income [s12, s13]

1. Tax = $206,583.50 + 37% x ($1,000,000 - $768,700) = **$292,164.50**.
2. Effective rate = $292,164.50 / $1,000,000 = **29.22%**.
3. Another $100,000: marginal 37% x $100,000 = **$37,000**.
Marginal (37%) is above effective (29.22%) because each rate applies only to income above its threshold.

4.3 Mortgage interest deduction

- **Interest is deductible; principal is not** (principal repays the loan, it is not an expense).
- Homeowners who itemize deduct interest on up to $750,000 of debt (made permanent in 2025). Investors deduct interest as an expense against rental income.
- **Tax saving = interest x marginal rate. After-tax interest rate = rate x (1 - marginal rate).**

Worked and practice [s15 to s17]

$2,000,000 IO loan at 10%, 37% bracket: interest $200,000; saving **$74,000**; after-tax rate 10% x 0.63 = **6.30%**. At a 12% bracket: 10% x 0.88 = 8.80%.
Practice: $300,000 IO at 12%, 37%: interest $36,000; saving **$13,320 a year**; after-tax rate **7.56%**. Each $1 of interest costs $0.63 after tax. Principal repayments save no tax.

4.4 Depreciation

- A deduction for wear on the building with **no cash spent**. **Depreciable basis = cost - land** (land does not wear out).
- Tax life: **27.5 years residential, 39 years nonresidential.** Straight-line deduction = depreciable basis / tax life (rate 1 / 27.5 = 3.64% a year).
- Active owners: rental losses offset wages only up to $25,000, phased out from $100,000 to $150,000 of income. Real estate professionals who materially participate are exempt.
- **At sale, depreciation taken is recaptured**: when price exceeds adjusted basis (cost - depreciation taken), that gain up to the depreciation taken is taxed at up to 25%.
- Before/after 1986: 15-year 175% declining balance vs 27.5-year straight line. PV of deductions at 10% on a $100,000 building: $56,791 vs $33,589; tax savings at the top rate then $28,395 (50%) vs $9,405 (28%). The 1986 act cut both the deductions and the rate.

Worked and practice [s19, s21, s22]

$1,000,000 apartment building (land excluded), 27.5 years: deduction **$36,364** a year; tax saving at 37% = **$13,455** a year.
Industrial building bought 10 years ago for $1.1m, land $100,000, 39-year life: **1.** ($1,100,000 - $100,000) / 39 = **$25,641 a year**. **2.** 10 x $25,641 = **$256,410**. **3.** Adjusted basis = $1,100,000 - $256,410 = **$843,590**. Only the building is depreciated; the $100,000 land stays at cost.

4.5 Capital gains and the three taxes at sale

Long-term rate, 2026	Single taxable income	Joint taxable income
0%	Up to $49,450	Up to $98,900
15%	To $545,500	To $613,700
20%	Over $545,500	Over $613,700

- Long term = held more than one year. Short-term gains are ordinary income. Plus 3.8% NIIT above $200,000 single / $250,000 joint. A $100,000 gain taxed at 20% instead of 37% saves $17,000.

Three taxes when a property is sold [s24]

Tax	Base	Rate
1. Income tax on the year's taxable operating income	Operating income	Ordinary, up to 37%
2. Depreciation recapture	Gain up to the depreciation taken	Up to 25%
3. Capital gains tax	Gain above original cost	0%, 15% or 20%

Exam trap

Recapture applies only when the price exceeds the adjusted basis. Below it, the sale is a loss: no recapture or capital gains tax.

A 1031 exchange defers taxes 2 and 3, not tax 1.

Taxes by sale price [s25]: $1,000,000 building, land excluded, held 10 years, adjusted basis $636,364

Price below $636,364: loss, no tax. $636,364 to $1,000,000: every dollar of gain is recapture at 25%. Above $1,000,000: recapture stops at 25% x $363,636 = **$90,909**; the rest is capital gain at 20%.

Examples: $800k = $40,909 recapture, $0 gain tax. $1,200k = $90,909 recapture + $40,000 capital gains. $1,400k = $90,909 + $80,000.

Practice: three taxes at sale [s26, s27]

Bought for $1.1m, $256,410 depreciation taken, sold for $1.5m. Operating income $50,000. Rates: income 37%, recapture 25%, capital gains 20%.

Income tax: $50,000 x 37% = **$18,500**. Recapture: $256,410 x 25% = **$64,103**. Capital gain: ($1,500,000 - $1,100,000) x 20% = **$80,000**. **Total $162,603.**

2. A 1031 exchange defers recapture and capital gains: **$144,103**. The $18,500 income tax is due either way.

4.6 The 1031 exchange

- **Why it exists:** swap investment real estate for more of it and defer tax on the gain. Owners sell and reinvest instead of holding to avoid tax, which adds liquidity.
- **Requirements:** real estate held for investment or business, for the same (no homes, no flips). **Identify the replacement within 45 days; close within 180 days.** A **qualified intermediary** holds the cash; the seller never touches it. Partnership interests do not qualify; tenancy-in-common shares can.
- **Boot is taxed now:** any cash the seller keeps, and less debt (repay a $4m loan, borrow $3m, and $1m is taxed like cash).
- **To defer it all:** buy equal or greater value and reinvest all of the cash.
- **Hold until death:** heirs' basis steps up and the deferred tax is never paid.

Worked example [s29]

Apartment bought 30 years ago for $1,000,000 (land $100,000). Depreciation taken $900,000; adjusted basis $100,000. Sale at $10,000,000: gain $9,900,000.

Recapture $900,000 x 25% = $225,000. Capital gain $9,000,000 x 20% = $1,800,000. **Deferred by buying a $15,000,000 building: $2,025,000.** New basis = $15,000,000 - $9,900,000 = **$5,100,000**.

Boot: keep $1,000,000 of cash and $1,000,000 of the gain is taxed now.

4.7 Entity choice, the 2025 tax act, bonus depreciation and Opportunity Zones

Entity	Taxed	Losses pass to owners?
C corporation	Twice: company, then shareholders	No
S corporation	Once, by the owners	Yes
Partnership or LLC	Once, by the owners	Yes
REIT	Once, if it pays out 90% of taxable income	No

Most private real estate uses LLCs and partnerships: one level of tax and depreciation losses flow to investors. REIT dividends are taxed to shareholders, mostly as ordinary income.

What the 2025 tax act changed (Public Law 119-21, July 4, 2025) [s31]

- **Bonus depreciation: 100%, permanent** (property acquired after January 19, 2025).
- **Opportunity Zones: permanent;** new zones from 2027.
- **LIHTC:** 12% more 9% credits from 2026; 4% credits now need 25% of cost financed with tax-exempt bonds, not 50%.
- **Mortgage interest:** $750,000 limit made permanent.
- **SALT:** cap was $10,000 since 2018; $40,000 in 2025, $40,400 in 2026, up 1% a year; phased down above $505,000 of income (2026); back to $10,000 in 2030.

Bonus depreciation [s32]

- Deduct **100% of short-lived property** (tax life of 20 years or less: appliances, carpet, fixtures, parking lots, landscaping) in Year 1. **Not the building** (27.5 or 39 years) and never land.
- **Cost segregation** moves building cost into short-lived property.
- It is a **deferral**: the extra deductions are recaptured at sale.

Worked: apartments bought for $7,000,000, land $1,500,000, 37% bracket

No study: $5,500,000 / 27.5 = $200,000 deducted Year 1; tax saved **$74,000**.
Study moves $1,100,000 to bonus property: $1,100,000 + $4,400,000 / 27.5 = **$1,260,000**. Tax saved **$466,200** in Year 1, **$392,200 more** than without the study.

Opportunity Zones [s33]

- Low-income census tracts nominated by governors. Invest a capital gain in an OZ fund **within 180 days**.
- For investments from 2027: **Defer** (tax due at sale or in 5 years, whichever first). **Reduce** (10% of that gain is never taxed; 30% in rural zones). **Exclude** (hold 10 years and the fund's own gain is tax-free).

Worked: $1,000,000 gain invested in 2027, 20% rate

Without the fund: tax of **$200,000 now**. With it: tax on $900,000 = **$180,000, five years later (2032)**. Fund sold 2037 for $2,500,000: the $1,500,000 gain is tax-free, saving **$300,000**.

4.8 Tax credits, AMI and LIHTC

Key idea

A credit cuts tax dollar for dollar. A deduction cuts tax by the deduction x the marginal rate. $10,000 credit: tax falls $10,000. $10,000 deduction at 37%: tax falls $3,700.

Historic tax credit (HTC) [s35]

- **20% of qualified rehabilitation costs, taken over 5 years.** $10m rehab = $2,000,000 of credits, $400,000 a year.
- Building on the National Register; work meets National Park Service standards; rehab must be substantial (more than the building's adjusted basis). Many states add their own credit.
- Harrison Square (old Cook County Hospital) raised $24m of HTC equity with no other government subsidy.

Low-income housing tax credit (LIHTC) [s36, s37]

- Credits each year for **10 years**: about **9%** of eligible cost (competitive) or **4%** (with tax-exempt bonds). States award them. Affordable for at least 30 years.

- **Income test:** at least **40% of units** for households at or below **60% of AMI**. **AMI** = area median income set by HUD per metro (half of families earn more, half less), set for a family of four, then adjusted for household size.
- **Rent limit = 30% of the income limit.**

Unit	Persons	Size factor	60% income limit	60% rent limit
Studio	1.0	70%	$42,000	$1,050
1 bedroom	1.5	75%	$45,000	$1,125
2 bedroom	3.0	90%	$54,000	$1,350
3 bedroom	4.5	104%	$62,400	$1,560

Lafayette FY2026 AMI = $100,000 for a family of four. Income limit = $100,000 x size factor x 60%. Rent limit = income limit x 30% / 12.

Blended AMI and income averaging [s38, s39]

- A unit's AMI level = the share of AMI it serves. RFP: rent / 100% AMI rent (a $1,500 one-bedroom / $1,875 = 80.00%). LIHTC: the income limit set for the unit, 20% to 80% of AMI.
- **Blended AMI = unit-weighted average AMI level.** Income averaging (since 2018): at least 40% of units affordable with limits **averaging 60% or less**. Higher rents at 80% pay for deeper discounts at 40%.

Worked

40 of 100 units affordable: 20 at 80%, 20 at 40%: (20 x 80% + 20 x 40%) / 40 = **60%, passes**. With 20 at 50% instead of 40%: (20 x 80% + 20 x 50%) / 40 = **65%, fails**.
RFP building (16 studios, 32 one-beds, 24 two-beds, 8 three-beds, 20 units at 60%): (16 x 74.29% + 32 x 80.00% + 24 x 84.44% + 8 x 92.31% + 20 x 60.00%) / 100 = **77.14%**. RFP counts every unit, reports (does not cap) it, and requires at least 20% of units at or below 60% of AMI.

Financing historic and LIHTC projects [s40]

- Credits are **not refundable** and these projects earn little taxable income, so the developer **sells** them. An investor buys **99.9% of the partnership** for cash and keeps the credits and the losses. The developer stays GP and buys out the investor later.
- **Break the rules and credits are recaptured:** 5 years for HTC; 15 years for LIHTC.

LIHTC options [s41, s42]

Cost $26.0m + $2.2m land. Sale in Year 15: $40m. Option 1 (no credits): all equity $28,200,000, market NOI $1,759,968 a year. Option 2 (LIHTC): investor pays $17,745,000 for credits; equity $10,455,000; NOI $717,928.
IRR: **Option 1 = 7.81%. Option 2 = 13.55%.** The credits replace $17,745,000 of equity and more than pay for the NOI that falls $1,042,040 a year.

Practice: tax credits [s43, s44]

Historic apartment building in Lafayette: $5m purchase + $8m renovation = $13m. Credits sell for $5m of equity; bank lends $7m IO at 5%. NOI $350,000; sale Year 15 for $9m.
Your equity = $13,000,000 - $5,000,000 - $7,000,000 = **$1,000,000**. Years 1 to 14: $350,000 - 5% x $7,000,000 = $0. Year 15: $9,000,000 - $7,000,000 = $2,000,000. **Levered IRR = 4.73%** (CF0 = -$1m, 14 years of $0, then +$2m). Without credits you would need $6,000,000 of equity.

Lesson 7 review [s45]

- Property taxes: rate x assessed value; Cook 10% or 25% x 3.03; abatement caps tax at the old value; TIF lump sum = PV of the increment.
- Income taxes: brackets apply to taxable income; marginal rate drives decisions; interest and depreciation cut tax; bonus = short-lived property, 100% in Year 1.
- Sale and entities: three taxes (income, recapture, capital gains); 1031 = swap and defer; boot is taxed now; pass-throughs taxed once.
- Credits, AMI, OZs: $1 of credit cuts $1 of tax; LIHTC = 40% of units at 60% of AMI; blended AMI = average AMI level; OZ = defer, reduce, exclude.

Part 5. Midterm Review practice problems, solved

Problem 1: apartment acquisition

Buy for **$7.5m**. Sell at the end of Year 3 to an outside investor at a **5.75% cap rate**, with **4% selling expenses**. Loan: 3-year, **$5m interest-only at 6%**, annual payments, **1 point** origination fee. Bank requires **max LTV 70%** and **min DSCR 1.25x**. NOI: Y1 $450,000; Y2 $463,500; Y3 $477,405; Y4 $491,727.

> **Key idea**
>
> **Assumption: sale price = next year's NOI (Year 4) / cap rate.** That is why the problem gives a Year 4 NOI. If your course convention differs (Year 3 NOI), the answers shift; check against your class notes.

A. Unlevered: net sale proceeds, Year 1 cash-on-cash, equity multiple, IRR

Item	Calculation	Result
Sale price, end of Year 3	$491,727 / 0.0575	$8,551,774
Selling expenses	4% x $8,551,774	$342,071
Net sale proceeds	$8,551,774 - $342,071	**$8,209,703**
Unlevered cash-on-cash, Y1	$450,000 / $7,500,000	**6.00%**

Year	0	1	2	3
Unlevered cash flow	-$7,500,000	$450,000	$463,500	$477,405 + $8,209,703 = $8,687,108

Equity multiple = ($450,000 + $463,500 + $8,687,108) / $7,500,000 = **1.28x**. **IRR** (IRR function on the four cash flows) = **9.06%**.

B. EBC of the loan if held to maturity; maximum loan under LTV and DSCR

- Fee = 1 point = 1% x $5,000,000 = $50,000. Interest = 6% x $5,000,000 = $300,000 a year.
- **EBC** = RATE(3, -300,000, 4,950,000, -5,000,000) = **6.38%.** It exceeds the 6.00% coupon because the borrower receives $4.95m but repays $5.0m.
- **Max loan, LTV:** 70% x $7,500,000 = **$5,250,000**.
- **Max loan, DSCR:** max debt service = $450,000 / 1.25 = $360,000. At 6% interest-only the loan can be $360,000 / 0.06 = **$6,000,000**.
- **Maximum loan = the smaller = $5,250,000. LTV binds.** The $5m loan passes both tests (LTV 66.7%, DSCR 450,000 / 300,000 = 1.50x).

C. Levered: required equity, Year 1 cash-on-cash, equity multiple, IRR

- **Required equity** = $7,500,000 - $5,000,000 + $50,000 fee = **$2,550,000**.

Year	0	1	2	3
NOI		$450,000	$463,500	$477,405
Interest		-$300,000	-$300,000	-$300,000
Net sale proceeds				$8,209,703
Loan repayment				-$5,000,000
Levered cash flow	-$2,550,000	$150,000	$163,500	$3,387,108

Levered cash-on-cash, Y1 = $150,000 / $2,550,000 = **5.88%**. **Equity multiple** = ($150,000 + $163,500 + $3,387,108) / $2,550,000 = **1.45x**. **IRR = 13.90%**.

Key idea

Read the result. Cash-on-cash falls (6.00% to 5.88%) because the loan's EBC (6.38%) is above the 6.00% going-in yield: negative leverage on cash. IRR rises (9.06% to 13.90%) because the loan's cost is below the unlevered IRR: positive leverage on return. Same pattern as Lesson 6, IBC 2.

Problem 2: office building with capped equity

$3.75m of equity and no more; $15m office building. Primary lender prohibits second mortgages. Two 5-year interest-only loans, annual payments, no fees: **A:** $11.25m at 7.5%. **B:** $9.75m at 6.5%, so you need **$1.5m of mezzanine**.

A. IBC of the additional $1.5m in Option A; the most you would pay for mezzanine

	Loan	Rate	Annual interest
Option A	$11,250,000	7.5%	$843,750
Option B senior	$9,750,000	6.5%	$633,750
Difference	$1,500,000		$210,000

IBC = $210,000 / $1,500,000 = **14.00%** (same as RATE(5, -210,000, 1,500,000, -1,500,000)). **The most you would pay for $1.5m of mezzanine is 14.00%.** Equity is capped, so the alternative to mezz is the bigger senior loan in A.

B. Mezzanine quoted at 11%: which option? Why does mezz exist when second mortgages are prohibited?

- Option B + mezz interest = $633,750 + 11% x $1,500,000 ($165,000) = **$798,750** a year on $11.25m, a combined cost of **7.10%** (vs 7.50% for A). It saves **$45,000** a year.
- 11% is below the 14% ceiling, so mezz is cheaper than the extra senior in A. **Select Option B.**
- **Why mezz is available:** a second mortgage is a lien on the **property**. Mezzanine is secured by a pledge of the **ownership interests** in the borrowing entity, not the building. The senior lender keeps its first mortgage, but if the mezz lender forecloses, the senior lender faces a new owner rather than a competing lien on its collateral.

Problem 3: tax savings from deductions

Married joint filer, **37% marginal rate**. Tax saving = deduction x marginal rate.

	Deduction	Tax saving at 37%
A. Interest on $900,000 10-yr IO loan at 6%	$900,000 x 6% = $54,000 a year	$54,000 x 0.37 = **$19,980 a year**
B. Depreciation on $7,500,000 apartment, land $1,450,000, 27.5-year life	($7,500,000 - $1,450,000) / 27.5 = $6,050,000 / 27.5 = $220,000 a year	$220,000 x 0.37 = **$81,400 a year**

- Interest saving = after-tax interest rate of 6% x 0.63 = 3.78% on the loan.
- Principal repayment saves no tax; land is not depreciated.
- Depreciation is a deferral: it is recaptured at sale (up to 25%).

Part 6. Rapid-fire self-test

Cover the answers and quiz yourself. Questions mirror the concept bullets on the Midterm Review sheet.

Lesson 5

#	Question	Answer
1	Name the steps of a market analysis.	Scan the environment; analyze the local market (demand); analyze the competition (supply).
2	What drives demand for multifamily rental? Industrial? Retail? Hotel?	Multifamily: households under 35, renter share, rent vs income, cost of owning. Industrial: manufacturing and logistics employment, freight volume. Retail: households and income in the trade area, traffic counts. Hotel: air passengers, conventions, tourism.
3	Which phase of development is riskiest and why?	The first (acquisition). Risk is highest when the least money is invested; required return ~40% falls to ~8% as risk burns off.
4	How does a land purchase option limit risk?	It gives the right, not the obligation, to buy at a set price by a set date. Loss is capped at the option price (plus legal fees) while due diligence and rezoning play out.
5	What affects the price of an option?	The length of the option period and the uncertainty about value; both raise it.
6	Define FAR, density, efficiency, parking ratio.	FAR = gross conditioned area / site area (parking excluded). Density = units / acres. Efficiency = net rentable / gross conditioned. Parking ratio = stalls / units.
7	Why was the $13.85m parking saving "not available" at Chauncey Square?	One level gives 241 stalls vs 390 required (0.46 ratio vs 0.75 minimum). The parking minimum is the binding constraint.
8	Vacancy 20% on 1,000,000 SF; occupied space rises 800,000 to 830,000 SF. Net absorption? New vacancy?	+30,000 SF; vacancy 17.0%.
9	Can vacancy rise in a year of positive net absorption?	Yes, if more space is delivered than absorbed.
10	Why is leasing activity not net absorption?	A tenant moving between buildings adds leasing but no occupied space.
11	What is a polycentric city and what is the evidence?	A city with several centers, each with its own bid-rent curve. Indianapolis: Keystone office rent ($24.89) is above Downtown ($24.22).
12	Why does a shopping center have one landlord?	A single owner collects the spillover between tenants in rent; separate owners each ignore the customers they send next door.
13	Income needed to afford $1,500 rent?	$1,500 x 12 / 0.30 = $60,000.
14	Demand funnel: 40,000 households, 40% renters, 35% qualified, 5% capture?	280 units.
15	Break-even rent vs market rent: what does failing mean?	The project fails the financial test: cut cost, add density, or walk.

Lesson 6

#	Question	Answer
16	How do construction lenders protect themselves?	Loan-to-cost cap, equity first, in-balance clause, credible take-out, draws against inspections, retainage and lien waivers.
17	Who pays for a cost overrun and why?	The developer (GP), through the completion guarantee and in-balance clause. The LP is not required to fund overruns.
18	List the developer's four guarantees and when each burns off.	Completion (at completion); carry (at stabilization); repayment, e.g. 25% of loan (at take-out); carve-outs (never).
19	Why does the construction lender size on cost and the permanent lender on income?	No income exists during construction; the permanent loan repays from stabilized NOI.
20	Why are the loan and interest reserve circular?	Reserve is a cost, loan = 70% of cost, interest depends on the loan. Iterate until they agree.
21	Amount due on $400,000 drawn at the start of each month for 15 months at 7.40%?	FV(7.40%/12, 15, -400,000, 0, 1) = $6,304,691; reserve = $304,691.
22	State the stacking rule.	Add a layer while its IBC is below the cost of the next source; stop at the binding constraint.
23	Where does mezzanine sit and what secures it?	Between senior debt and equity. Secured by the ownership interests in the borrowing entity, not the building.
24	Maximum you would pay for mezzanine?	The IBC of the additional senior debt it replaces (or your cost of equity if equity is available, whichever is lower).
25	Why can you get mezz when the senior lender bans second mortgages?	Mezz is not a lien on the property; it is a pledge of equity interests.
26	GP vs LP: capital, control, liability, guarantees.	GP: small capital share, runs project, unlimited liability (held via LLC), gives lender guarantees. LP: most capital, votes on major decisions only, liability limited to capital, no guarantees.
27	Preferred equity vs LP equity with a preferred return.	Preferred equity: fixed return, redemption date, can take control if unpaid, behaves like expensive debt. LP equity: priority inside a shared upside, no deadline, behaves like equity.
28	What is a true hurdle?	An IRR test: a balance grows at the pref rate and falls with each LP distribution; the split changes when it reaches zero.
29	Define promote.	The GP's share above its share of capital. The LP's 80% is never a promote.
30	How does phasing limit risk? What does it cost?	Guarantees are capped and burn off phase by phase. Costs: lost scale economies, cost growth, option premiums, market risk, building next to an occupied phase. It moves risk; it does not remove it.
31	What is the cross-collateral trap?	Pledging phase 1 to fund phase 2 ties phase 2's risk back to phase 1.
32	What is the real option in the BCBS tower and in the land option?	Pay a premium now for the right, not the obligation, to grow (tower) or buy

		(land).

Lesson 7

#	Question	Answer
33	Abatement vs TIF.	Abatement freezes the tax at the pre-improvement value; benefit arrives slowly as lower taxes. TIF gives the developer the PV of the future tax increment as an up-front lump sum; the city borrows against the increment.
34	TIF lump sum formula?	PV of the increment (after-improvement tax minus before-improvement tax) over the TIF term at the city's rate.
35	Effective vs marginal tax rate: which drives decisions?	Marginal. Effective = total tax / income.
36	What does a $300,000 12% IO loan save a 37% filer?	$36,000 x 37% = $13,320 a year; after-tax rate 7.56%.
37	Depreciation on $1.1m building, $100k land, 39 years?	$25,641 a year.
38	What is adjusted basis and when does recapture apply?	Cost minus depreciation taken. Recapture applies only when price exceeds adjusted basis; taxed up to 25%.
39	Three taxes at sale and which a 1031 defers.	Income tax on operating income; depreciation recapture; capital gains. A 1031 defers recapture and capital gains, not income tax.
40	1031 timing and mechanics.	Identify replacement within 45 days; close within 180; qualified intermediary holds cash; like-kind real estate for investment or business.
41	What is boot?	Cash kept or net debt relief; taxed now.
42	Which entity types pay one level of tax and pass through losses?	S corporation, partnership, LLC. C corp pays twice; REIT pays once if it distributes 90% of taxable income but does not pass losses.
43	What did the 2025 act do to bonus depreciation, OZs, LIHTC?	100% bonus permanent for short-lived property; OZs permanent with new zones from 2027; 12% more 9% credits from 2026 and 4% credits need 25% (not 50%) bond financing.
44	OZ benefit structure?	Defer the gain (to sale or 5 years), reduce it by 10% (30% rural), exclude the fund's own gain after 10 years.
45	Credit vs deduction.	Credit cuts tax dollar for dollar; deduction cuts tax by deduction x marginal rate.
46	LIHTC basics.	Credits for 10 years; ~9% or ~4% of eligible cost; at least 40% of units at or below 60% of AMI; rent limit 30% of income limit; affordable 30+ years; sold to an investor for cash; recapture over 15 years.
47	Historic tax credit?	20% of qualified rehab costs over 5 years; National Register building; substantial rehab; recapture over 5 years.

48	Studio 60% AMI rent limit if AMI is $100,000 and size factor 70%?	$100,000 x 0.70 x 0.60 = $42,000 income limit; $42,000 x 0.30 / 12 = $1,050.

Notes and flags

- **Lesson 4 is not in the files.** EBC/IBC/LTV/DSCR definitions come from how Lessons 5 and 6 use them.
- **Problem 1 sale-price convention** (forward NOI / cap rate) is my reading of the problem; confirm against class.
- **Slide inconsistency:** L5 s7 says "about 8% once stabilized" but the chart shows 10% in the stabilization row; L6 splits it as lease-up 10% / stabilized 8%.
- **My answers vs slide answers:** the L5 s22 pairing decision and the L5 s28 exercise have no answer on the slides. Mine are labeled. All other slide answers were re-computed and match, except for a rounding note: L6 s15 states the lender yield as 8.88% a year (a nominal rate: monthly yield x 12); the effective annual yield is about 9.25%.
- **Figures in the decks are illustrative or sourced to the decks' citations.** I did not re-pull any live market data (Indianapolis vacancy, rents, tax brackets); those are as printed on the slides.